

Stage 1

Multi-Signal Exposure Mapping



S₁: Embedding Similarity



S₂: Min-K%++



S₄: Self-Critique

RRF $\rightarrow e_{ij} \in [0, 1]$

e_{ij}



Stage 2

Contamination-Augmented IRT

2PL + γ_k model

Bayesian NUTS

θ_j, γ_k



Stage 3

Dose-Response Profiling

Algorithm 1

classification

Immune

Collapse

V-recovery

Benefit